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Professional Self-Assessment

Introduction

Completing the Computer Science program at Southern New Hampshire University has been one of the most transformative experiences of my academic and professional journey. Through each course and project, I have developed both technical expertise and the confidence to apply those skills in real-world scenarios. The *Daily Drop* app, which served as my primary artifact for enhancement, represents the culmination of my growth as a software engineer. It showcases my ability to design, build, and refine applications that are functional, user-centered, and aligned with professional software-engineering standards. This self-assessment reflects on my progression throughout the program, my development of key competencies, and how these experiences have prepared me for a career in software engineering.

Professional Growth and Development

Throughout my coursework, I have gained hands-on experience with multiple programming languages, development environments, and frameworks. Courses in algorithms, databases, and full-stack development challenged me to think critically and problem-solve systematically. Early in the program, I focused on learning syntax and logic; by the end, I was designing and implementing complete software systems.

The *Daily Drop* app exemplifies this growth. What began as a simple weight-tracking concept evolved into a polished mobile application featuring data persistence, visual hierarchy, and SMS functionality. Implementing SQLite integration and developing an interactive UI through

Android Studio helped strengthen my understanding of database design and user-interface structure. This project also taught me the value of iteration each enhancement cycle gave me the opportunity to refactor code, optimize queries, and apply design principles that improved performance and maintainability.

Collaboration and Communication

Another area of growth has been collaboration and communication. Many of my SNHU projects required working within simulated development teams, participating in peer discussions, and documenting progress clearly. Learning to write concise code comments, maintain version control through GitHub, and communicate design decisions effectively has prepared me for the collaborative nature of the software-engineering field.

During the development of Daily Drop, I treated my GitHub repository as a professional workspace organizing commits, documenting code enhancements, and structuring directories for clarity. These habits mirror real-world industry practices and demonstrate my ability to work in environments that value transparency, version control, and effective communication among team members.

Application of Core Computer Science Principles

The most rewarding part of this program was learning how to integrate the three major computer-science domains software design, algorithms and data structures, and databases into cohesive applications. In *Daily Drop*, I applied object-oriented design to structure the app logically, ensuring each class served a single, clear purpose. Algorithmic efficiency was considered in how data was stored and retrieved, reducing load times and ensuring responsiveness on mobile devices.

My database enhancement reinforced my understanding of CRUD operations, indexing, and query optimization within SQLite. Meanwhile, my work in software-engineering courses emphasized modular design and user-centered principles. Together, these components demonstrated my ability to design secure, scalable, and efficient systems that adhere to modern development standards.

Ethics, Security, and Professional Responsibility

Beyond technical proficiency, the program emphasized ethics and security two critical areas in today's software landscape. Throughout my coursework, I learned how to evaluate code for vulnerabilities, implement encryption when dealing with sensitive data, and follow best practices for user authentication and privacy. In *Daily Drop*, I implemented proper user-input validation and ensured that sensitive information was handled securely within the app's database.

Understanding these responsibilities has strengthened my perspective as a developer. I now recognize that writing secure code and maintaining data integrity are not optional tasks but essential components of professional ethics in computer science.

Integration of Coursework and Artifacts

The three enhancements software design, algorithms and data structures, and databases collectively showcase the progression of my skills. Each milestone built upon the previous one, creating a final artifact that reflects the breadth and depth of my education. The iterative process of refining my code, documenting enhancements, and conducting a formal code review mirrored the lifecycle of professional software projects.

This process also taught me the importance of self-evaluation. Reviewing earlier versions of my work allowed me to identify weaknesses in my logic and architecture, make targeted

improvements, and measure tangible growth over time. This reflective approach is something I intend to carry into my career as a continuous learner and developer.

Conclusion

As I prepare to graduate and enter the professional field, I feel confident that the knowledge and experiences I gained through the Computer Science program have equipped me to succeed as a software engineer. My journey from understanding basic programming concepts to creating a full-featured mobile application demonstrates perseverance, adaptability, and dedication to my craft.

The *Daily Drop* app, my enhanced artifacts, and my professional ePortfolio collectively illustrate my readiness to contribute to a development team, design efficient systems, and uphold ethical standards in software creation. This capstone experience has not only validated my technical abilities but has also shaped me into a thoughtful, purpose-driven engineer prepared to grow in the evolving world of technology.